ADAM TANG

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EDUCATION

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering and Computer Sciences, Minor in Bioengineering — *Expected June 2029*

- **Relevant Coursework**: CS61A - Python, Scheme, SQL | Math 53 - Multivariable Calculus | Engineering 183A - Innovation Entrepreneurship | Bioengineering 26

Phillips Exeter Academy | High School Diploma — Exeter, NH

- 1600 SAT | **Coursework:** Multivariable Calculus | Linear Algebra | Accel. Bio/Chem/Physics — *September 2021 - June 2025*

WORK/ENGINEERING EXPERIENCE

Eldaeon (Skydeck Startup) — Berkeley, CA
Electrical/Mechanical Engineering Intern — *December 2025 - Present*

- Power mapped and designed DPOL system architecture for Eldaeon's passive radar, reducing power consumption by **30.2%** and ensuring system safety under all component energy constraints.

CalSol (UC Berkeley Solar Racing Team) — Berkeley, CA
Solar Engineer — *September 2025 - Present*

- Optimized solar array layout and bypass diode configuration for Gen 11 vehicle; improved photovoltaic efficiency by **12%**, contributing to vehicle's hardware subsystem optimization
- Engineered integration of wiring and lightening holes into carbon fiber shell and improved maintenance access

MUREX Exeter (MATE Marine Robotics Competition Team) — Exeter, NH
Mechanical Lead — *November 2021 - June 2024*

- Led design and fabrication of custom 6-DOF underwater ROV and 7-DOF manipulator arm, + independent buoyancy module
- Cut manufacturing costs by **88%** through in-house component design using Onshape + CAM workflows
- Team broke **10 industry records**, including creating the world's **smallest and most affordable network switch**
- Team placed **6th** globally at MATE 2024 and earned awards for Best Engineering Presentation, Poster, and Perfect CAD

Research: Microplastic Pollution in Rivers (Catholic University of America, CEE Dept) — Washington DC
First Author, Researcher — *June 2022 - August 2024*

- Simulated vertical transport of microplastics using flume hydraulics; generated **250+** data points used in dispersion modeling
- Developed drone sampling protocol to access **3+ previously unreachable zones**, enabling full environmental diagnostics
- **First-author** research paper selected for WaterSciCon 2024; project awarded by NASA, EPA, GENIUS Olympiad, and ISEF

Formula 1 in Schools — Exeter, NH
Mechanical Lead — *September 2022 - June 2023*

- Designed F1 car using Fusion 360/ANSYS simulations; optimized aerodynamics via iterative testing in wind tunnels
- Achieved **1st place at State Finals 2023** through performance optimization and weight reduction

TECHNICAL PROJECTS

PolySynth X - Expressive MPE Keyboard & Embedded Synth Platform (in progress) | *November 2025 -* — Berkeley, CA

- Led complete R&D lifecycle from hall sensor physics and mechanical prototyping to firmware optimization and live testing
- Developed and optimized compliant key mechanism for per-key pitch modulation, DSP engine w/ Daisy seed microcontroller

Low-Cost Modular Biodiversity Sensing Module (in progress) | *September 2025 -* — Berkeley, CA

- Developing solar-powered wildlife sensing nodes with camera, acoustic, and thermal inputs; lead on **embedded hardware**
- Implementing **Wi-Fi HaLow telemetry** to enable remote, solar-powered sync; prototyping **embedded board with edge AI**
- Collaborating cross-functionally on **AI/ML species ID pipeline** for autonomous biodiversity classification in field

Fully Automated Aquaponics System | *December 2023 - June 2025* — Exeter, NH

- Designed and built closed-loop aquaponics system with **automated pH, ammonia, and temperature sensing**
- Engineered integrated biofiltration and nutrient cycling using siphon effect to optimize water quality and plant growth
- Acquired iterative feedback from national/international experts (UNH, Victory Aquaponics, Island School (Bahamas)); led cross-functional ops with faculty/staff to deploy system as live food + edtech solution

LEADERSHIP

Innovation Towards the Future (NGO) — Potomac, MD
Founder & President — *September 2020 - June 2025*

- Represented the **USA** as youth delegate at the 28th UN Climate Change Conference (COP28) in Dubai, chosen from **5k applicant NGOs**; participated in formal global climate negotiations and policy formulation
- Led NGO to be admitted to the UNFCCC (UN Framework Convention on Climate Change)
- Forged **3 corporate partnerships** to distribute eco-friendly bags/cups to communities in need
- Launched and promoted global No Single-Use Plastic Campaign through national and international network connections

SKILLS

Programming: Python, Java, SQL, MATLAB, TypeScript/JavaScript (Next.js, React), AI/ML (learning)
Engineering Tools: ANSYS, Fusion 360, Onshape, FreeCAD, Arduino, Raspberry Pi, KiCad, 3D Printing
Systems/Concepts: System Performance Optimization, Embedded Systems, Failure Mode Analysis, Python Scripting, Mechatronics
Soft Skills: Public Speaking, Cross-Functional Collaboration, Human-Centered Design, Stakeholder Engagement, Mandarin